

EMERSON, MB, Canada

WMO: 715600

Lat: 49.0014N

Lon: 97.2417W

Elev: 794

StdP: 14.28

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.7	-18.4	-30.3	1.0	-22.2	-26.0	1.3	-18.2	36.0	6.5	32.1	5.6	7.1	320	0.679

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.0	86.8	70.5	83.8	69.1	81.1	67.3	74.4	83.0	72.0	80.4	69.8	78.0	12.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.6	120.2	80.2	69.0	109.8	77.4	66.7	101.1	74.7	38.6	83.1	36.3	80.6	34.3	78.2	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.9	25.3	22.3	DB	-28.8	92.7	5.8	1.7	-33.0	93.9	-36.4	94.8	-39.6	95.8	-43.9	97.0	(4)
(5)			WB	-29.1	78.6	5.7	2.2	-33.2	80.2	-36.6	81.5	-39.8	82.7	-44.0	84.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.0	5.2	8.5	22.9	40.6	53.9	63.4	68.1	65.8	57.4	42.7	26.9	11.0		
	DBStd	24.73	14.30	13.75	13.75	10.80	9.09	6.36	5.56	6.08	8.35	9.15	11.37	13.78		
	HDD50	5961	1388	1161	843	311	65	1	0	0	26	264	694	1208		
	HDD65	9787	1853	1581	1306	731	358	101	32	65	252	692	1143	1673		
	CDD50	1959	0	0	2	30	185	405	561	489	248	38	1	0		
	CDD65	310	0	0	0	0	13	55	128	88	25	1	0	0		
	CDH74	3191	0	0	0	10	225	542	1175	922	296	21	0	0		
	CDH80	821	0	0	0	0	63	123	308	254	70	3	0	0		
	WSAvg	10.4	11.5	11.0	11.1	11.5	11.5	9.6	7.9	8.0	9.7	11.0	10.9	11.0		
	PrecAvg	19.2	0.5	0.4	0.7	1.1	2.4	3.6	3.0	2.6	2.1	1.4	0.7	0.6		
	PrecMax	29.8	1.6	1.5	3.0	3.3	6.6	9.0	8.5	8.1	6.1	5.0	3.9	1.3		
(15) Precipitation	PrecMin	10.7	0.0	0.0	0.0	0.0	0.5	0.6	0.5	0.5	0.1	0.0	0.0	0.0		
	PrecStd	4.3	0.4	0.4	0.7	0.9	1.5	2.0	1.8	1.8	1.4	1.1	0.8	0.3		
	0.4%	DB	37.8	41.6	61.7	74.8	87.3	87.5	89.7	90.8	86.4	76.9	58.2	40.9		
	MCWB	34.9	38.8	53.2	54.3	62.9	69.7	74.4	73.0	67.2	60.1	49.4	36.1			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2%	DB	33.8	36.6	50.9	68.8	79.9	83.5	86.3	86.0	81.6	68.8	51.5	35.6		
	MCWB	32.3	34.7	45.3	52.8	59.6	67.3	72.6	70.8	65.9	56.9	45.2	33.2			
	5%	DB	29.6	33.3	44.3	63.7	75.0	80.1	83.5	82.4	77.1	62.8	46.7	32.6		
	MCWB	28.0	32.0	40.5	49.9	57.5	65.8	70.8	68.6	63.2	53.4	41.5	30.9			
	10%	DB	25.6	29.4	38.8	58.6	70.4	76.6	80.6	79.2	72.6	57.5	41.5	29.4		
	MCWB	24.5	28.0	36.4	47.7	56.2	64.1	69.1	66.6	61.2	50.0	38.1	28.0			
	0.4%	WB	35.1	39.7	54.1	58.1	67.2	75.0	77.8	77.0	72.2	63.1	51.4	36.5		
	MCDB	36.9	41.7	58.9	69.5	78.1	83.1	86.1	85.6	82.2	72.4	56.7	40.0			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	32.1	34.9	46.1	54.0	63.4	70.9	75.1	73.1	68.4	58.3	45.8	33.6		
	MCDB	33.5	36.2	51.1	65.3	73.1	79.0	83.3	83.0	77.1	67.5	50.6	35.4			
	5%	WB	28.1	32.1	40.7	51.0	60.7	68.1	73.0	70.6	65.4	54.1	42.1	30.9		
	MCDB	29.4	33.3	44.2	61.6	69.6	76.4	80.9	79.3	73.3	61.6	46.1	32.5			
	10%	WB	24.5	27.8	36.7	48.2	58.3	65.6	70.7	68.3	62.6	50.4	38.1	28.0		
	MCDB	25.6	29.2	39.0	57.8	67.4	73.7	78.5	76.4	70.8	56.8	41.5	29.3			
	5% DB	MDBR	16.1	17.4	16.7	20.2	22.1	19.8	21.0	22.9	22.0	17.7	15.1	14.8		
	MCDBR	17.0	14.7	20.0	29.3	29.5	25.0	24.0	26.5	28.5	26.9	21.8	17.1			
(36) Mean Daily Temperature Range	MCWBR	15.6	13.1	14.8	17.1	14.5	12.7	13.2	13.6	15.3	16.4	16.0	15.0			
	5% WB	MCDDBR	16.8	14.3	19.3	26.3	23.1	20.5	21.5	22.9	23.5	24.8	20.3	16.4		
	MCWBR	15.5	12.8	14.8	16.8	13.5	12.2	13.2	13.1	14.1	15.8	15.3	14.5			
	taub	0.273	0.300	0.305	0.353	0.370	0.383	0.386	0.381	0.347	0.305	0.275	0.257			
(40) Clear-Sky Solar Irradiance	taud	2.272	2.204	2.386	2.351	2.366	2.354	2.355	2.366	2.457	2.542	2.492	2.386			
	Ebn at Noon	249	267	289	284	282	278	275	272	271	268	253	244			
	Edh at Noon	23	31	31	36	37	38	37	35	29	23	18	18			
	RadAvg	468	860	1248	1561	1737	1866	1988	1637	1195	722	411	323			
(44) All-Sky Solar Radiation	RadStd	50	61	115	134	127	143	82	106	84	101	55	37			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	-1.57	-1.85	N/A	N/A	N/A	N/A
Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page